

ActInf Livestream #031.2 ~ "Non-equilibrium Thermodynamics and the Fre

Livestream | 1:52:10

hello everybody and welcome to the
active inference lab today it's october
26th 2021

and we're in active live stream number
31.2

welcome to the actimf lab we are a
participatory online lab that is
communicating learning and practicing
applied active inference you can find us
at the links here on this slide
this is a recorded an archived live
stream so please provide us with
feedback so that we can improve our work
all backgrounds and perspectives are
welcome here and we'll be following good
video etiquette for live streams
we're here in our second discussion
group discussion for 31 towards the end
of october so also happy halloween
and this is our jumping off point to
many ideas so looking forward to this
cool discussion

the goal is going to be to learn and
discuss

this paper by patricia palacios and
matteo colombo both of whom are with us
today so thanks so much to the authors
for joining and

feel free to address what you can while
you're here and then just leave whenever
works for you

and in this discussion we can
take it many different ways because so
many topics came up in the dot zero and
dot one but we have all of the

formalisms some big questions a lot of
topics that we raised in the 31.1 video
as well

so we can begin with an introduction and
warm-up section we'll have the
non-authors introduce themselves and
then just say hello and then we'll pass
to the two authors both of whom are
joining for the first time to a stream
so again thanks for
joining and we look forward to the
discussion

so i'm daniel i'm a researcher in
california

and to just add something i'm excited
about matteo and i were just discussing
right before we started some of the
sociology meets history and philosophy
of science

and oh yeah who could forget about the
actual topics themselves so there's just
some cool intersections that we'll be
able to talk about

i'll pass it to dave

i'm retired from information technology

i'm in the cool dry mountains of the
northern philippines

and now that i don't have to work for a
living i'm getting really busy on
learning cybernetics which is when one
of the things i did back in college and
having a great time is fired up about
having a complexity weekend ramp up
starting tomorrow

sojourn stephen

we can't hear you steven but you're
unmuted so maybe your audio
selection is odd

but let's um
welcome steven back so figure it out and
let's go to um patricia first welcome
and happy to hear your introduction
hello well thank you for for the
discussion for
for the interest in discussing our paper
so i'm an assistant professor in
salzburg i'm originally from chile
and i work in the philosophy of physics
philosophy of science
and now i'm working more and more in the
philosophy of the complexity science
and i have done some work on going to
physics
and
well i'm interested in general in the in
the statistical mechanical applications
to other disciplines um well
statistical mechanical models
in economics biology
and well chemistry although i haven't
explored that already
yet i'm interested on that
and
with mateo was my i did my first
exploration in the free energy principle
but um just to be clear i'm very
interested in the application of
statistical mechanics to other
disciplines even if we appear to be
critical to that in this in this paper
thank you matteo
thank you very much for uh
the kind invitation and also for the
spotlight uh on our paper completely
undeserved
my name is matteo colombo

i'm based in tilburg in the netherlands
uh as my accent might suggest i'm
italian i'm originally from milan and
mostly my work lies in the philosophy of
cognitive science i have a special
interest in computational modeling
especially reinforcement learning and
bayesian approaches
uh to brain function
and
you know it's been fun
to collaborate with patricia at this
project which um you know combined a
bunch of interest already had and helped
me to learn something new
great well maybe we could just start
with
some of the context of the paper like
how did both of you
intersect towards this very
interesting view on fep and active so
were you studying active inference and
then wanted to integrate it with
statistical physics or vice versa i
think we heard some of the seeds of that
but it'd be good to clarify
yeah if you want yes okay so
i think
patricia and i started to chat about
this uh two or three years ago at a
conference uh problem uh so
my background in uh is in uh quantum
neuroscience and uh
philosophy
mainly so masters involved and during my
work at edinboro
as a phd student i was actually
co-supervised by a person

peggy serrier
who is an expert in
psychophysics
and then she moved slowly uh to apply
tools from
basian modeling and reinforcement
learning modeling to psychiatry and so
my and my other supervisor was
andy clark and so was a good
um
say uh
environment for me you know to
learn about active inference i think
those were the early days of this
approach
and started to are asking questions
basically about the approach in terms of
actual applications a very limited
experience
i suppose that
the only uh serious modeling experience
said was during my uh phd
where with another collaborator we
started to model
uh things in uh
abroad bayesian framework
um this paper came in to be i think
mainly because of my own interest in
bayesian modeling and patricia's own
interest
which she just mentioned in statistical
physics approaches
to economics and to other sciences and
so we started to ask hey
in terms of scientific representation
that's the focus of our paper focus is
not so much
as also in

written in the paper on the metaphysics
of you know active inference but more
about japanese language this formalism
the scientific representation and let's
ask what kinds of opportunities and
challenges that this formalism
presents when it comes to applying it
outside
its say
original field as in you know statistics
statistical physics
and so we started to chat and uh with a
back and forth
for a while and then we
um
say
started to put into focus
various opportunities and challenges in
applying
say
concepts from
thermodynamics and statistical physics
to make sense and understand
organisms and life science phenomena and
what
you know we found very interesting is
this kind of trade-off between the
generality of it that is it seems that
the machinery of the free energy
principle or active inference uh you
know we might have a discussion about
the difference if any between the two
how this machinery allows allows you to
cover at least a model
many types of systems at least allegedly
but with a sacrifice in a plausibility
uh what we call a realism in the paper
and again by itself is trade-off we

don't think there is anything wrong with
uh you know
going for maximal generality in an
approach that might cover you know or at
least allegedly from cells
up to you know cultural phenomena
um and that's our paper in a nutshell
that is how these trade-offs might
depend on a choice of
language or formalism to make sense of
allegedly any
thing and again the thing good
is one of the key terms i think
that the maxwell and you know before him
carfreestone uh put on the map here and
i think it's a helpful way also to look
at our contribution again
in a nutshell i think the
framework of thinggood
uh might miss out
the most interesting properties of the
kind of thing
if there are things um say organisms
happen to be
thanks for that patricia anything to add
i'll suggest just um to to repeat what
what matteo said so my interest in the
free energy principle came from my
interest in the application of
statistical mechanics to all the
disciplines so that's why we started
discussing this and that's why i got so
excited about this about this research
and i agree with the summary of the
paper that mateo just just did
this thread of grounding in fundamental
physics ideas and then trade-offs or
opportunities and challenges really

became clear so
maybe one
general opening question
what does it mean for a non-statistical
physics idea to be grounded in
statistical physics like we're
intrinsically talking about some
interdisciplinary or transdisciplinary
or applied scenario so we're leaving
that thing that field of statistical
physics
what baggage good and bad do we take
with us and then is it an advantage is
it a weakness what does it actually mean
for a non-physics area to be grounded in
physics
yeah thank you for the questions that's
one of the questions i have motivated my
research in the last years
uh i don't think there is a unique
answer to that so the more i do research
on the applications of statistical
mechanics to other discipline the more i
understand that it varies so it depends
on how it is used how how statistical
mechanics is used
and it depends also on the case studies
and disciplines and the specific models
so i think that's why it's worth to look
at specific case studies to specific
applications of free of
statistical mechanics to the disciplines
in order to answer that question like on
a case by case basis
i think in some cases the application of
statistical mechanics is well justified
especially when the idealizations are
well justified in some other cases as i

would say tend to think that it's the case of the free energy principle the application of statistical mechanistic lack justification so we are not saying or we are not concluding in the you know paper that that cannot be i know that it doesn't make sense to propose a principle that is based on statistical mechanics but we just think that there is much more work to be done in the justification of especially of the idealizations that are involved when we use statistical mechanics to formulate a principle like the free energy principle classic philosophy where do we air or miss when we idealize make ideas of or generalize specific systems or things um and what is the thing concept or the system concept in either patricia or mateo in either of your areas like if you mentioned a thing or a system what would that mean in to the the disciplines that you're working with and then how is it being used differently in fep and active and then i see steve my dad you want to answer this sure um i mean in a sense your question is about metaphysics uh and the question is what's the thing and what uh in what way we can make sense of a

thing good uh with you know a formalism

like the formalism

markov blankets for instance or the

formalism of uh

statistical

mechanics

um

just to turn the point slightly in a

different direction if you may

um

i suppose the uh question about thing

good and the role of idolizations on

which you know patricia just

commented on

is not just boiling down to the to the

you know conclusion that ah that's

difficult let's move on uh to to apply

these tools um

to a certain system

uh the challenge we put on the table

here is

concerning the generality of it

uh the generality of the act inference

approach

uh you know there is a excellent work uh

by uh philipp schwantenberg and others

using act inference models

uh in uh market decision processes with

humans you know fitting decisions in

reinforcement learning or decision tasks

with active inference models and again

this is just one example of a productive

use of the active inference

approach which you know resonates with

regular approaches like bayesian

modeling or reinforcement learning

modeling

what

uh we're taking issue with is
look you're making claims
about a maximum scope
okay maximum scope kind of application
of this approach
and to sensibly apply
you know models
at any scale you know any scale even
outside the life sciences you know
having to do with economies
or thing good certain assumptions must
be in place and these assumptions you
know we provide a number of
examples which all involve the
availability of
large data sets okay that can constrain
uh say the application of the model and
also background knowledge about
the target this is the target system and
the thing you want to understand
and especially you know bridges
conceptual bridges between parameters
within your model
and features of the actual systems you
want to
represent so these are the three i think
challenges that
we put on the map for again the
large
scope or very general scope
some understanding of active inference
or the free energy principle so this is
a bit of topic so i didn't really answer
your question about thing but at least
these are
motivations we had
to write this paper as in
making claims about maximal scope come

with
commitments and these commitments at
least in the case of
many or most biological systems do not
look uh satisfied or plausible
thank you
in a way it does
oh wait maybe a little bit mateo
but um in a way it does
address this question about our models
as things and their relationship to
other things perhaps the target of
modeling so patricia if you want to add
anything
otherwise we can go to stephen
just to stress some of the things that
matteo said because i hear the session
offline i hear the session of last week
and i i i saw that um
maxwell was
concerned or pointing out that we
didn't take the cases that the
applications of the free energy
principle to some systems in biology
seriously
and but the thing is that our claim we
did take them seriously and we mentioned
them in the paper
but that doesn't contradict uh our
claim because our claim wasn't that
the
statistical mechanics cannot or the
language of dynamical systems cannot be
used to model biological systems our
claim was about the generality of this
of this principle so basically we were
concerned or where we
were casting doubt on the possibility of

formulating a general principle like the
free energy principle that presupposes
that all
relevant biological systems can be
modeled as random dynamical systems so
just to emphasize that our claim was our
concerns with the free energy principle
had to do with generality of this
principle and not with the possibility
of modeling uh
some biological systems in that way
using the language of random dynamical
system

thank you so stephen and then anyone in
the live chat or anyone with a raised
hand here

yeah thanks this is very interesting
hopefully you can hear me okay um
i'm curious um i know that in the the
the reference you make to the paper that
is um

life as we know it so
there there's that question about um
suggesting that life or biological
self-organization is an inevitable and
emerging property of any ergodic random
dynamical system that possesses a markov
blanket

and i know that
in some ways i i i sense that what
they've done to try and
deflate that claim from 2013 is maybe to
go down the mathematical route
i s i sense that you're saying they
need to maybe also go down the physics
route would that
would that be something
that you would say like in terms of how

that um

that statement

which has to some extent been couched
over the last few years

um are you saying that these you know

the the physics aspects

um haven't been addressed as much as the
statistical aspects

i can say a couple of things and then i

leave it to you patricia yes very very

quickly um

so um

one

uh line

uh of criticism about our paper is that

you know

[Music]

the active inference approach has been

evolving over time and so we're making

claims about ergodicity and

which are just obsolete now because

uh you know from the coast hostile

paper uh that assumption seems to be

dropped

uh and replaced with uh the the

assumption of stationarity

but then the question actually two

questions one question is

why should we believe that any uh

life science phenomenon any organism is

stationary this what is it what is the

justification for that assumption

and the second uh

observation

before i leave it to patricia is that

even within the paper you know that the

constitutional paper which is you know

um

theoretically uh very rich

they

frame the contribution to biology in
terms of a metaphor

explicitly early on they're saying look

what we're seeing here is just a

metaphorical uh representation

and so in a sense that's what we're

being arguing for that is sure you can

gain this maximal generality but at the

cost of uh making assumptions either

ergodicity or structurality now which

might might not apply to biological

systems

thank you patricia

yeah so

to to answer stephen's concern i would

say yes so one of the things that we

wanted to point out is that

we need to focus on the on the physical

foundations of the free energy principle

a little bit more than only the

statistical foundations of the free

energy principle so we believe we don't

claim to be the only ones that focus on

on that but we believe that there's lack

of literature in that sense and we

wanted to contribute to uh to to that to

fill that gap what we call feel that god

uh so that's

that's the short answer and then um

about again the discussion that you had

last last week

uh you saw maxwell i think as well was

complaining that we focus too much on

the older formulations of the free

energy principle instead of the

enumerations and he basically said that

ergodicity doesn't play any role it's not even mentioned in the new formulations of the free energy principle

i don't have a biological background or background in neuroscience and i haven't worked for a long time in the free energy principle but i do have a background in physics

and i am a little bit puzzled with the idea that ergodicity doesn't play any role in the new formulations of the free energy principle especially because they explicitly allude to

the existence of weakly mixing random dynamical systems and to my knowledge mixing systems require ergodicity so implied ergodicity so i'm really not sure and i haven't seen

explicitly how can you drop the assumption of eliciting the free energy principle so

i think that if they if some authors working on the free energy principle could clarify to what extent or why you don't need their basicity anymore i think that would be helpful so we would be grateful for that we wouldn't

consider that as a failure in our project so we would consider it as a success so if our paper leads to more clarity in the free energy literature that would be a successful paper for us and and i think the fact that bill edal also some paper that was quoted in this discussion last week also

allude to the older formulations of the free energy principle just shows

maybe a confusion in the literature between older formulations and new formulations so i think again it would be very helpful for people interested in this at least to to have more clarity about what has changed in this new framework in these new formulations of the free energy principle and to have a more consistent framework that one can work with because we were interested in working with the legitimate framework we weren't against the free energy principle and then tried to look for the weak for the weakest argument to attack them so that was never our strategy i don't have anything but i am not married against the free energy principle i don't have anything a priori against crystal so the same with matteo so we are just generally interested in in this application because it can be actually potentially useful or because it has been so uh it has caught so much attention in the in biology so that's why we are interested in this in the first place not with the idea of attacking the free energy principle so again if we could have better understanding of why the new formulations don't require ludicity or about what are the key assumptions in the new formulations of the free energy principle i think that would save us time and we would be happy to to look at that so i i think again

i i agree that we focus mostly on the older versions of the free energy principle but i think it was partially because the papers were clear and not because we wanted to attack the weaker

papers in the same like the fact that they appeal again to the existence of weakly mixing random dynamical systems suggested for me that they still are using ergodicity

and apart from that ergodicity was not the only argument that we were presenting in the paper so we also mentioned phase spaces and attractors and the new formulations still require phase spaces because they need some probability and they uh they also consider or model homo aesthetics or non-equilibrium steady states as uh in terms of attractors

and that again falls into our criticism because we were pointing out independently of the ergodicity assumption we were pointing out that it's very hard to model in most systems biological system even physical system perhaps um

stationary non-equilibrium stationary states in terms of attractors because attractors implies some asymptotic results and it's not easy to get these asymptotic results

and so basically that was our criticism so we were by no means only focusing on ergodicity even if ergodicity doesn't play any role in nervous formulations in in new formulations which is not clear to

me

but even so i would say the other points
still remain and we would like to to
have a clear answer to that even even if
that means that we have to i don't know
say that we were wrong in the end but um
i insist so if our paper leads to more
clarity in the literature that we will
be fine we will consider it as success
thanks for that and welcome blue so i'll
just give one comment and then stephen
so

absolutely about the evolution and
changes it's a non-stationary or maybe
there's a higher level of a stationarity
to the literature but it's one reason
why blue and i and maybe you if anyone
listening wants to contribute seriously
to this are very interested in tracking
the

nomenclature and which letters are used
to refer to internal and external states
and which assumptions are implicit or
explicit how does that play out with the
evolution of literature in this area so
that's like something that we'll
hopefully will be able to point to
maybe it's going to be version 1.4 in
this paper or you know version 1.4 the
the english gambit we can name them like
chess openings

okay stephen

yeah i think you make a good point i
mean i think also i i do agree in the de
costa paper stationarity has really
given us another way in to some sort of
state that gets revisited it doesn't
necessarily

take out a good city maybe adds another
potential so
there's this this question
doesn't it can't just be answered from a
math perspective like i think it's
there's this physics
um reality and maybe this is where
non-equilibrium chemistry starts to come
in or that's a new field um
and i i think that your your points are
are useful in clarifying
some of those questions i suppose the
question that also comes up which is
maybe less common in physics
is
how much can
science how much can
um a kind of noisy
version of ergodicity
be enough
and at what point you know how many
beers is it before i fall over you know
at what point does um
does the active influences process come
down and i i don't actually think that's
necessarily been
brought in but i don't know if physics
knows quite how to handle that question
yes how stringently do specific
realizations or models especially if
they're being evaluated on utility
how stringently must they fulfill
some of the underlying frameworks like
we talked about how the t-test is built
around equal variances unless you're
using an unequal variance version
and of course those will never be
precisely realized but then

how close to being precisely realized
must or should things be
um matteo or patricia any thoughts
oh otherwise the the comment that came
to mind was patricia you mentioned um
and it was in your 2018 paper i think
that we looked at in the dot zero
about how there's the asymptotic
characteristics and the convergence time
so that was very interesting um just
that there was a there was it's sort of
this
attractor but there was more to the
attractor than just where it was
so maybe
unpack that and then how did you see
that grafting into or relating to the
um comments on fvp and active
yeah exactly
sorry i have to leave now thank
you you know
it's been fun uh
i'm very sorry you know it lasted only
30 minutes because
i saw so blue joined and i had answers
to a couple of their questions in my
previous sessions about dynamic
equilibrium and
why we use internal
uh states uh as active states
well we can do 31.3
literally anytime we can do it in a year
so that's why we do decimal notation any
time it's right just let us know and
you're always welcome back for a guest
stream presenting on current work so
good luck with your teaching
responsibilities and see you soon

too generously church hello everybody

and thanks for paying attention

cheers to patricia

okay

yeah so

yeah you were mentioning my 2018 paper

thank you er yeah so that

i think the the question of asymptotics

is related to the question of

approximation science and i am

interested in both questions

and this is basically what i have done

independently of the free energy

principle in the in my research in the

last in the last year so basically what

i what i pointed out about asymptotics

in the paper that you were mentioning

is that um

in order to make a model realistic a

model that takes a synthetic

into account in order to make that model

realistic

we not only have to pay attention to the

solutions that you get in the limit when

you take a limit you also have to pay

attention to how fast that limit

converges

in that time that can be considered

relaxation time or that

time that it takes that that the limits

needs

in order to to

or that we need in order to achieve that

limit um

needs to be realistic too

so you don't only you

you don't only have to pay attention to

the values that you take that you that

you obtain when you take a limit you
also have to
to pay attention to the
to
um
the rate in which you approach that
limit
so that will also help you to understand
how realistic is the model that you are
that you are using
how would that tie to approximation
science
uh it depends again
some some models like on the models that
you're analyzing in the science that you
are looking at
so if you are
investigating models that use limits
then there is a very clear understanding
of approximation because uh limits give
you sort of a neat
understanding of the approximations that
you are using
but approximations are not always uh in
the form of limits sometimes
approximations can be in the form of
similarity and then
saying to what extent that model is
realistic it can be a little bit more
obscure
so when your model implies limit then
you might have a clear way of
understanding to what extent your model
is close to to the to the target that
you are trying to mode
very
interesting so
one

thought that's um come up
in a few different contexts is um it's
almost like a vitalism 1.0 and a
vitalism 2.0
and the 1.0 was like you know is there
the elan vital is there something more
to it than
chemistry in biology and you know the
synthesis
ex vitro or x vivo should i say of
vitamin c for some reason it's pointed
to as an example of like demonstrating
that biological molecules could arise
through chemical synthesis
and therefore maybe more than just
vitamin c but maybe dna could also be
made in the test tube
and now we're in a area where we're
asking okay
given that grounding of biology in
physics and chemistry 1.0 how do we go
from that sort of
equilibrium
into homeostasis and allostasis and more
advanced forms of
predictive
higher level
behavior
from systems that are also
composed of less strategic components
so
um
there's going to be just like there were
challenges going from the
physics of a membrane into understanding
how biological systems deploy membranes
now we're moving up a level into the
cybernetic or into the anticipatory

space and asking how those very same biological systems which we've deflated away to be chemical and physical now will there be another level of deflation or is this a bubble that's a little harder to pop

so

steven and then patricia or anyone else with thoughts

yeah this is a good point about

it being a bit harder to pop

um because there's a lot of uh once you get into the chemistry there's a lot of um

there's a lot of hidden states let's put it like that so there's that assumption that you pass through often maybe a transition state when you go from reactants to products but it's going from equilibrium to equilibrium even the transition state is looked at in terms of its energy

accessibility

but

that's again a kind of a stable state so this question about something circling around non-equilibrium or far from equilibrium

um mixing states which is why i think you can't i think they say ergodicity can't be or something along those lines can't be ignored

uh certainly once you get into anything which is more in the liquid

uh

contexts

um so the question comes up then is when

you've got these
attractors or these far from equilibrium
states which are not necessarily
energetically favored
alone they may be somehow entropically
favored or somehow complexity favored
through
these attractors you know is that
something around how catalysts are
working
um in the more sort of if you were to go
in at a more detailed level of modelling
um so i wonder how

[Music]

your thoughts are around the sort of
catalysis processes that might be
happening in biological systems and
how
you know how do these far from
equilibrium states
facilitate
these
these abilities to
minimize free
energy if you have any thoughts not
patricia otherwise blue
do you wanna
i would leave that to blue we have to
work more on the free energy principle
i don't know i i think uh the assumption
that i've worked more on the free energy
principle is maybe kind of
i don't know
not
not right i don't really have an answer
don't let the cat ears deceive you
[Laughter]
but

cool questions um
there's many other avenues we can go but
patricia what is your current
research
or what do you see following your work
now that it
came out like what where do you go from
that state update
to your next questions if they're
related or not it's all good
yeah actually i'm thinking about
applying for a project
uh which it doesn't have to do with free
energy principle but it does have to do
with applications of statistical
mechanics to biology
so i have been working for a few years i
will probably i'm so i just um finish a
book on emergency reduction in physics
so i have been working on a concept of
emergency reduction for physical systems
especially thinking about critical
phenomena
and what i want to do now is to see
whether that notion of emergence and
reduction that i suggest also works for
biological systems but i will focus on
biological systems that can be modeled
as critical phenomena which means can be
model using the physics of phase
transitions
so this is what i what i have in mind
now and i i i intuitively think so my
binary uh hypothesis is that
the same concept doesn't work for
biological systems and well that one
needs to distinguish between them but i
think it would be useful to look at

specifically at systems that can be modeled as critical phenomena to see this analogy maybe between biological systems and physical systems so that's maybe the reason why i will focus especially on those systems that can be according to to to many physicists b model and biologists be modeled as a critical phenomenon interesting in fact criticality is uh perhaps again the language analysis will reveal in the future but it's not a term that gets brought up in the active and fep area that much it's more apt to describe assist although in fristen's work on neural systems there actually is a lot of it and so there's actually so much in the spm and dynamical modeling literature that has um not made it quite over the hill but i think that there's more to be said we'll look forward to that book stephen yeah just that's an interesting point and they're making um can i just ask so is it almost like you think there's a lot of um a lot of computational processes that can happen purely through phase transitions and statistical processes which is not quite the same as an inference

process but it is it may be more like
the sort of thing you see when
molds um

find their way to food and things like
that so you sort of saying there's a
kind of a physics
explanation for some things which
doesn't require

an influence process
and that could do a lot of the heavy
lifting

yeah so i'm i will focus on on on models
that use the mathematics of phase
transitions with variations so you never
use exactly the same mathematics you
have to you have to make change
depending on the case study that you're
investigating

but i will i will focus on on models
that um that use the mathematics of
phase transitions to understand
biological phenomena

uh
an example would be [___] behavior for
example so there are plenty of models
that use the icing model or something
like the icing model uh to understand
falcon behavior which is that behavior
of birds that seem to be emerging with
respect to the behavior of individual
birds

and in other case studies like melding
protein melding that can be modeled as a
phase transition

so i'm going to i'm going to analyze
plenty of case studies of research that
have been done already in physics or in
biology

and the idea would be to understand what kind of notion of emergence do you find in biological systems and the question is whether that is similar to the notion of emergence that appears in in physical systems and as i said i intuitively think that it's not the same but since i will look at similar case studies i think that can show the difference between biological systems and physical systems quite quite clearly

[Music]

there's one thing i was curious then is it like because this is interesting when i'm talking to people who've got maybe more of a chemistry background emergence is more the emergence of interactions between things right molecules and stuff and then you get these kind of and then you've got this more process based kind of inference dynamical one so are you you sort of seen the way of emergence being able to do quite a lot from the kind of perspective of things interacting with each other based on rules rather than non-linear dynamical approach both actually so you find both in phase transitions that way that's why phase transitions in physics have been considered as the prototype example of emergent behavior because you find

non-linearity you find um
long range correlations
and well the interactions in most
systems are local but you still have
these long-range correlations
that require using some fun segments
like the randomization group methods for
example

so you have some notion of collectivity
there in in phase transitions in physics
not because of long-range interactions
but because of long-range correlations
so it appears as if the
interaction between between uh distant
particles would be would be long-range
it is not but it appears like that
because you have long-range correlations
so i will look at similar cases in
biology that have been modeled already
using physics

and then i will try to answer whether
that notion of emergence in terms of
non-linearity and so on
what i suggest it's emergence emergent
behavior in physics can also be applied
to biology

and i think well but this is this is
preliminary so this is not these are not
the results of my research because i
will be doing that for the next years
but a preliminary thought
is um

you will see a difference especially
with top-down causation that might
happen in biology that doesn't happen in
physics but this is just preliminary so
again we try not to say too much about
this because i haven't done the pro i

just have done the research for the project but i haven't done the project so

you can you can wait for this result thank you

blue

so that's super interesting um and again something that like i'm very interested in is this um idea of emergence and when you're you're studying phase transitions or criticality in physics like for example just to from a liquid to a gas you know it's easy to see on a graph this kind of inflection point of temperature where the phase transition occurs

um

and i just am wondering what kind of metrics you're using to quantify emergence um in your future research in biological systems

uh that depends again

sorry that many many times my is the same but that depends on the i'm not using a specific metric because i'm not proposing as a general framework i'm just studying like different case studies and depending on the case study on the kind of phase transition that i'm studying is the the metric that is attached to it

but i'm not i'm not suggesting a general formal framework that has a metric in itself so i'm i'm studying like case by case

so can you maybe provide some examples or you don't want to say yet too much in biology

yeah the [___] behavior so there are different different models there i can you can you can send me an email if you want and i can address you to some this literature but there are plenty of models of funky behavior right that use that use models that are already so that sort of migrate models from physics so there's some of them use a slightly revised version of the of the icing model it makes me think about how uh the sort of initial target system let's just say a solid to a liquid phase transition and then there also are potentially critical dynamics related to information and decision making which comes into play in active inference where you could have uh a uh non-linear relationship between information coming in and then it's in policy mode a and then there's a phase transition to policy b and so um it's kind of like again that vitalism 1.0 2.0 there's going to be a lower bar a phase transition to talk about well the protein folding in the cell is going to be a special surrounding that facilitates it to do special kind of transitions physically but then how do we think about higher order behavioral properties

especially when they exhibit
some
tantalizing similarity or analogy or
metaphor like
we've been discussing but can we just
take those same models and think that
it's going to be as clean all the way up
um blue and then stephen
so in the flocking model
um for example like and you're talking
about downward causation uh
and something as something that you wish
to study and so i'm sure that you've
maybe read um the information theory of
individuality paper by um david pakkauer
and that group at the santa fe institute
so there's um
a point in which
you know this like group of birds that's
like flying around you know kind of
chaotically
um assembles and forms like a flock and
and um flies together in in like a kind
of patterned way and so like i'm
interested in in terms of criticality
at what point is
does downward causation occur right and
and is that the point that then it stops
being like the the ensemble becomes a
group a unit that's functional like you
know because there's it's like that's a
good kind of way to look at um emergence
but how do you find that point
in in which there's the bi-directional
information flow so this so that's
that's maybe a better way to phrase my
question
yeah so i assign this great to discuss

this with you but as i said i haven't
done the whole research i am i'm happy
to discuss my intuition here with with
all of you

but that's it yeah

i so one of the hypotheses that i have
is that you might find something like
that welcome down

downward conversation in biological
systems i'm not sure of flocking
behavior would be the place to find the
workout session but

i'm thinking about downward constant
work station in terms of selection
and it might be that you need some high
order

properties uh so that that sorry that
high order properties explain some
fitness

and that in that way that calls somehow
all the way down

that um

at the microscale at the micro level you
observe some behavior as a result of the
fitness that is uh sort of coarse grain
or or or higher level

so i'm thinking about that workout
station in the in those
terms

but you might find other notions of
downward causation there and i might
found them in doing the research
now in the flocking behavior i'm not
sure but you might think that the
the

the ability of many of many birds to to
go in a particular to to do a particular
pattern has some some

fitness associated to it and if so that
fitness which is fitness
of a group might explain the individual
behavior so if that's correct then you
might have a notion of downward
consensus that you don't find
necessarily in physics but uh this is
just pretty these are preliminary
intuitions
thank you blue
so like going towards like
along those lines um and going back
maybe to the attractor
so so when there is like a the emergence
of a group does does that like reign all
of the individuals in the group
like closer to a new attractor state or
closer together like tighter like a
tighter cluster around an attractive
state or is it like the new attractor
state is an emergent property of the
group or or how do those two fit
together like in terms of fitness
um and you know in biological fitness
you have to be able to maintain this
kind of non-equilibrium steady state
density
so in terms of fitness of a group or
cluster and biological systems
is that
like a new attractor or is there yeah or
just what are your intuitions maybe
about that
yeah so i i wouldn't i wouldn't try to
describe the notion of of emergence here
in terms of attractors but it might be
useful in the end
but i would like to uh to

have a notion of emergence that is that
doesn't rely necessarily on attractions
because i think that
finding attractors is is a hard task in
biology and some of these models don't
work with attractors anyway so i don't i
don't i don't think the notion of
emergence needs to be associated
necessarily to attract maybe in some
cases it would be useful to understand
emergency in that sense but i wouldn't
think that it's necessary to understand
emergency incentives for
attracting
and i agree with you but but um
like the coupling of emergence maybe to
fitness
to me like i link fitness to an
attractor and so it's it's hard for me
to separate then if if
fitness is linked to attractor but
fitness is
is not linked to um or is also linked to
emergence then that that draws a line
for me between
emergence and detractor if that makes
sense yes so that's why i will work with
biologists on this project i'm not a
biologist myself but i i i think that
fitness can be defined also independent
on the notion of attractors
and i will try to take those notions
cool we will try to adapt those
definitions
so dave just because you haven't asked a
question yet then stephen
yeah one of the reasons that uh looking
for a track spending too much time

looking for attractors associated with
them with emergence is
among the things that happens when you
get emergent behavior is there's a more
differentiated
set of interfaces with the outer world
and because of that there's less
duplication and less
resemblance among the
subsystems already within
the overall system
cool thank you stephen
yeah that's a good point around the
challenge of identifying that attract
especially if you've got this
multi-scale nested i mean so you've got
attractors that are only present for
nanoseconds
you know even if they're there and
there's millions and billions of them
you know you and then on top of that
you've so i can imagine that it could be
really useful and i i i'm curious um
how
you see that you know the freezing of
ice
um and the way that it you know water
becomes more dense and actually
about two degrees then starts to get
less dense
and as it approaches freezing and then
you sort of hit this phase transition
um and i think there's a real use in
that also being learned about for active
inference
because although an active influence
doesn't happen directly in the field of
anything to do with entropy there's this

common misconception of neg entropy as
if you actually get a negative form of
entropy
which is kind of in the
in the kind of common usage but there's
no such thing directly as a neg entropy
if there's a there's a negative there's
a change in entropy
you know there can be a variation in
entropic but it's not
a stable state in the same way that you
get with energy but that that
um i think that can be a misconception
that um
sort of creeps in quite a lot but i'd be
curious if um i
you know um
effectively you know there's there's
there's a value
of entropy that goes down spooky value
of entropy which no one quite knows
what's going on with the entropy we know
there's a value for entropy that
entropy goes
changes in from ice to water
but you know even
richard feynman used to say well no one
quite knows what's going on there and
maybe this is
something that you're trying to get into
is what's going on in those phase
transitions
and these entropic changes which
doesn't have the benefit like you say of
some a gentle
driver of the free energy principle to
make it happen so i don't know what your
thoughts are but how that might even how

that might be useful
to help people who aren't as heavily
into physics just think about the idea
of
change and entropy because i think it's
it's going to become of something that
more and more people are using
outside of physics
if you have any
dots otherwise just the maximum density
is 4 c for water
and it's super super slight but a very
interesting pattern
okay
we can turn to well if anyone wants to
raise their hand um or
give a thought
um
stephen you said like that the neg
entropy as like it might not exist that
was one thing and there's no such thing
as entropy
well you know yeah i also wanted to
sorry about about that
i
yeah i also wanted to to stress that in
equilibrium phase transitions in physics
you don't use the notion of negative
energy of course you just use entropy
and standard statistical mechanics plus
three normalization group methods
so that my
might be
an asymmetry with some of the case
studies that i analyzed in biology but
yeah we'll see
and one of the fundamental tradeoffs
discussed in the paper um was this

generality versus the plausibility and
the realism
do attractors even exist or is it just
as if
a system is moving towards something but
does something that
is not realized exists
and so this kind of uh takes us into
talking about
uh whether we're describing the system
as it is or rather situating the system
in a broader space of how it could exist
and that was one of the fundamental
points of the paper about the challenges
of discovering the appropriate phase
space for a biological
system
so
um
it's just cool because these are
not
knock down punches
for fep and active
nor are they trivial
they're in that
optimally informative or a confusing
space you know between zero and one
probability
where
it does matter
and uh where philosophy comes into play
and
where the the specifics of the situation
come into play we can't just simply make
um
arguments without reference to a
specific
model

but then are we trapped to just
describing specific models specific
papers or are we able to
go beyond that ever
i'm just looking through um if anyone
wants to raise their hand or any yeah
about that sorry
uh so
we do
raise a concern about the existence of
attractors
uh of course attractors don't exist in
nature yeah because you need you need
taking infinite limits in order to
define attractors in a model and you
cannot so infinities are not realistic
right so we cannot take
systems don't don't take an infinite
amount or processes don't take anything
out of time and
uh you cannot repeat an iteration in
renormalization group transformation
infinitely many times in a
in a realistic way of course these are
all idealizations but in principle i
wanted to stress that we don't have
we don't have a problem with that so of
course any model that uses attractors
wouldn't be
realistic in a way so you will be using
an idealization that's true
but we don't have a problem with that if
the idealization can be justified i
think in many cases especially for
example in
in
in the treatment of phase transitions in
in physics where you use uh limits

there's a way of justifying those limits
and justifying the existence of
attractors and therefore
even if you miss some realism in the
model of your of your physical system
since that idealization is justified
that that is plausible so that that
model is plausible and it's legitimate
what we are what we are
worried about in the paper is that it
might happen that sometimes uh the
models that well they try to model
many systems in terms of attractions are
not realistic they cannot justify the
idealizations that are being made in
order to model many plenty of systems in
terms of attractors
so i just wanted to clarify that because
we don't disagree in principle with the
idea of of using models that appeal to
attractors even if attractions don't
exist in reality our problem is well
with the possibility of finding
attractors in the first place or finding
asymptotic solutions
and the second is about the possibility
of justifying the idealizations that are
involved when you use
attractors or when you appeal to
attractors
yes and like many other
arguments in biology it's a little bit
like the anthropic principle or fallacy
depending on which side you're on it's
like
you mentioned in the paper you have to
prove the existence of the attractor as
well as as we discuss the time to

conversions and then justify why certain attractors

denote homeostatic states and one biologist's answer is just those are the living systems if the attractor weren't a homeostatic state or a high fitness state

that is not going to be an ant you see around for long and so we can just say that if there is an attractor and indeed it's as if there's an attractor for persistent systems that's the adaptive one period and the second that it's not that way

the system fails to exist so that's the sort of circular

but often impenetrable tautology of selection and persistence of systems blue

so that's a nice uh transition perhaps into the question that i've had about um dynamic equilibrium versus a non-equilibrium steady state density so i don't know if you want to flip to that side daniel but um there there's you know several quotes in the paper where um

the

the i'm not sure exactly what's referred to um as a dynamic equilibrium i'm assuming that it's uh

homeostasis

but um

i just was wondering why because a dynamic equilibrium is um a reversible chemical reaction and there's nothing about homeostasis that is i mean there may be some things

but not everything about maintaining
homeostasis

is has to do with reversibility um so i
just was um wondering

uh all biological systems actively
maintain a dynamic equilibrium with
their environment and what is meant by
that

this was a quote from the
paper

oh wait you're muted patricia but if you
have any thoughts

yeah so sorry i i was missing the screen
somehow disappeared everything

uh yeah blue so actually unfortunately
matteo wanted to answer that

in a in a

in a neat way uh basically we took this
from from freestone himself so he refers
in many places and he had the quotations
and so and i don't have that at the
moment i can try to look at it but

in a so you might be right it's not the
right way of of describing the

homostatic states but i think it's a

mistake that that freestone himself does

i think so this is basically what what

matteo wanted to to say and he had these
quotations in the papers where he does
that

so

if you want to email him he will he will
address you to those words but not no so
not so much

in our paper depends on that notion

actually so that might be a mistake that

we make i think because we need to give
freestyle

but not so much depends on that so
basically our criticisms have to do with
the idea of modeling

um

all biological states in terms of random
dynamical systems

in understanding homostatic states in
terms of dynamical attractors as they
are described in the paper

but

we don't rely so much on the notion of
dynamic equilibrium so that might be
just a loose way of

of understanding uh homostatic states so
it's referring to homeostasis though not
a dynamic testing okay yes yes yes

homeostasis yes yes

thank you stephen

i mean this could be an interesting
area for the for the ontological work
that's happening in the lab because
this question of

you know is it

dynamic equilibrium i think like blue
was saying there can

can be that kind of backwards and
forwards of which there's maybe a bit
more of an arrow in one direction and
then you've maybe got

the word equilibrium as opposed to
non-equilibrium

and then you've got this so he has this
term dynamical

systems you know

um

and um

as opposed to system dynamics which is
the kind of the way that say chemistry

like equilibrium work is more system
dynamics for dynamical systems
um and you've got this interesting thing
because you've got this non-linear
dynamical work happening at the sort of
the the the
physics level and this emerging property
level and then at some point
um
as it appears from the organism's
perspective
we might start turning and using the
language of dynamic equilibrium
ecologically you know or ecological
equilibrium
and this this shifting
the way that the
ontologies are present which may be true
because as things persist and interacted
with as entities
um they they appear to be
you know
in
a different type of dynamic but we are
seeing from
the active inference that that dynamic
may
in itself be more dynamical than we
perceive
so i don't know i just thought i'd
mention
um i don't know if we need an answer to
that
or maybe daniel you've got a thought but
there's something about
um
this shift between biology and
physics at the sort of

at this sort of very um
molecular stroke um you know theoretical
level
and then you've got the idea of what
that means for biology and then you've
got the idea of what that means for
biological organisms
um
and maybe this is something that we've
talked about this pluralistic ontology
that needs to be
feasible in the field because
like say it depends i think that was
a very good point it does depend on
where your
models are based and grounded
thanks
mateo has some very interesting work on
pluralism in biology and the philosophy
of science so i'd recommend people look
there
now
contingent explanations are not
necessarily plural one can be an
absolutist and say it depends on the
particulars and there's only one correct
story so it's an open question to what
extent pluralism and particularity will
coexist
and agreed that
at the high level
as we
mix metaphors take ideas across
different fields
we have to be aware of what is gained
and lost what are the challenges and
opportunities of applying
terms from mathematics to physics

and from physics to biology and taking
us all the way up and down that xkcd
comic that we were um looking at earlier
one

question area patricia that i had was
about

action in the loop

so

uh

maybe there's two parts to it the first
is just we often hear about fep as being
akin to a principle of least action

so just

from a physics background

what does it mean to be a principle of
least action or what would it mean to be
like a principle of least action

and then um

how

does it uh change the formalisms or not
when we're integrating control
of behavior kind of control theory
formalisms or cybernetics

into

ensembl modeling

because the actions of molecules are
implicit in the phase transition of
the ensemble of water but what happens
when we actually have a control
parameter

in our equations

like policy

okay thank you so first of all i did
find the papers where where frison uses
the notion of dynamic equilibrium so for
for blue

and he uses a synonym of of homeostasis
but i i cannot use the chat can you oh

it's on the left side uh the bottom left

it says that enter a nickname to use the
chat you can pick up whatever you want
okay

it was kind of possible okay

call me trim tab

okay

okay it was so i thought it was it was
sort of a password so you just wanted to
allow everyone to write in the chat it
doesn't require a child name but

okay

good

these are the papers blue so you can
take a look at this so yeah

but as i said not so not so much in our
paper it depends on that notion of that
so you might disagree with that lack of
record there

but yeah and don't blame us for that we
just okay

nice that was it interestingly not in a
peer review paper not that it
explains everything but he may be
speaking maybe yeah informally in this
aon piece

exactly i think i think it's just uh
yeah a more loose way of talking about
it and we also use it because it's
useful then when you speak about
dynamical attractors so if you have a
notion of dynamic equilibrium it's just
uh useful but of course we were we
weren't thinking about the the the the
proper definition of dynamical
equilibrium

so good now just return the action
question yeah okay

okay so yeah

again so we are here so sorry that i
excuse myself of summarizing the work
that frison did

uh but here we are just summarizing so
we don't give a proper definition of
principle of list action we're just
relying on on friston's own definition
and he basically in that paper of 2012
he needs the principle of of least
action in order to formulate the free
energy principles so the principle of
least action is a step
an intermediate step in order to
formulate the free energy principle
nothing more

then i would be more careful in talking
about principles of list action in in
the fa in phase transitions in physics i
think

you might

not even require any principle of list
action there you might talk about those
uh ground states as principle as uh sort
of

relying on a principle of less action
but i would be more careful about uh
exporting that notion to to the physics
of phase transition

okay

what's that kept up

yes so that's the first part about the
principle of least action and then just
is there um something that is changed
markedly when we introduce
these control theoretic
formalisms like policy selection
into physical

physically inspired or based
models
that entail activity like the thermal
motion of particles but are not
approached from a control theoretic or
behavioral regulatory perspective
are you thinking about social systems
for example
perhaps different kinds of systems but
um mainly just a
uh a particular system
uh in the way that uh free energy
principle for a particular physics for
students 2019 work like the particle is
the blanket and the internal states
that's engaging in active inference
let's say
and so it
is implementing inference on policy it's
selecting policy it's behaving and so
is that sort of policy selection latent
within
non-controlled theoretic
activity-based physics models
again like the thermal motion of
molecules or when we introduce policy
inference
does that change some of the
qualities of the system
i i think so it's an interesting
question i would need to think about
that a about it more carefully but my
intuition is that it does it changes the
the way uh i still understand uh phase
transitions in physics
um
i think that requires changing formalism
and again it requires uh important well

making important difference in your in
your model with respect to to how you
model phase transitions in physics
thanks

i agree something changes and it's not
for linear time discussion necessarily
to unpack but these are the kinds of
questions that we're getting at
absolutely so blue

yes yes i think

so um i just want to reference the um
articles that you posted in in the chat
so the am thing they did say dynamic
equilibrium but like i like don't
trust like op-ed science writers to
make direct uh correct quotations um and
in the ramstead paper that was cited in
your paper the markov blankets of life
it specifies that dynamic equilibrium
specifically is not

the state of homeostasis it says the
very existence of living systems can
therefore be construed as a process of
boundary conservation where the boundary
of a system is its markov blanket that
means that the dependencies induced by
the presence of a markov blanket are
what keep the system far removed from
thermodynamical equilibrium not to be
confused with dynamic equilibrium so i
mean specifically like there it's not
dynamic equilibrium it's it's far from
thermodynamic equilibrium but but still
dynamic equilibrium is is not not a part
of that and so that's why that reference
was very um confusing to me and like
they on paper is like yeah
like yeah and then we made the same

mistake that that he made without
correcting it
in the way that
it can be so i i agree we were talking
loosely there and we were what we were
having in mind was the notion of
dynamical attractors so that's why we
found uh dynamic equilibrium very very
like a a nice term but but we might be
talking loosely there but as i said so
that might be a mistake
but um not so much in our paper depends
on that notion
well so and the notion of a tractor in
the
in the state of homeostasis is
specifically a far from equilibrium
removed
space that a biological system exists in
and so it's just kind of um
yeah confusing to me i thought
interesting um
you know it's a challenge to say enough
to identify
what we're talking about but especially
when we're
crossing disciplinary
boundaries
to know which
pieces of baggage make it through one
border or another like dynamical
could merely just mean involving time
but
it also brings on a lot more meaning in
different areas
um
it's a great yes so we talk we talk in
the paper about the possibility of uh

attractors like that change well one

attractor following another so we

understood that as a loosely as a notion

of dynamic equilibrium but you are

not using the term rigorously probably

here

let's turn in the um sort of home

stretch of however long we want to speak

to the

short and concise conclusion

so

you you

in the last paragraph brought up this

really interesting point

about the fep as a definition

for a system

and so um

i wondered

what else could it be or might be what

might we want a framework to be other

than a definition like what else

in the counter factuals of the way that

this paragraph could have been written

what else could the fep or another

framework be other than a definition

is that synonymous or complementary with

a description

with

what other pieces come into play here

can you can you repeat the question i

don't know if i understood it um if the

fep

uh

is not a definition what else could it

be

or other than a definition what do we

want for a full

pledged model principle

what would make it a principle i think
uh
something that can be can explain
life or it can explain um
yeah why why why uh
biological systems uh
apparently violate the second law why do
they stay in that
far from equilibrium steady state
i think uh many
defenders or advocates of the
applicators of the free energy principle
understand this sort of as a principle
that explained that has explanatory
power
that explains the the the
not only the existence even the
existence of living systems but
especially why those
living systems
stay
uh
far from
from
equilibrium or from from maxwell
boltzmann equilibrium
for a long time in a in a steady state
great
point so
yeah that's smart
and explanation which of course goes
beyond or is complementary to definition
explanation is one of the key
terms and themes in the history of the
philosophy of science
and
since an explanation is evaluated
subjectively or inter-subjectively like

was that an adequate explanation
that's a statement about the
relationship between
a perceiver and the explanation itself
and so it uh dumps the entire mess back
into the lap of almost a
psycho-historical
perspective on science because
good explanation is about the
enculturation of the person evaluating
how good that explanation is so steven
and then blue
yeah i was just um looking up regarding
dynamical and dynamics so
dynamical the
adjective means relating to the study of
dynamics
so it says a dynamic system is a system
exhibiting continual change
so a dynamic system is continual change
a dynamical system is a system
relating to the study of the dynamics
so a dynamical system is like a system
working with
the dynamics of
a system and
dynamic itself just talks about the
actual fact that there's change
happening
so that's kind of um
that's kind of interesting and be
interesting as well whether you get
dynamical things
or there has to be systems you know like
the the the
the the is there
that may be where there's some sort of
interesting play between

uh phase transitions as well like maybe
there's some
there's a dynamic going on is there a
dynamic that is second order to the
first dynamic in some way so that that's
kind of interesting yes
things and processes
and static
versus dynamic and then
loose
but acceptable in many cases use of
language
and um
again the text modeling
will be
superlative when we can establish where
these adjectives and nouns and verbs
are being used appropriately versus
inappropriately
um or at the very least
suggesting more precision or less
precision than
uh warranted
so blue
oh and dave has a great comment there
historical
examined by historians using founded
methods versus historic which is like
about the event itself like it was a
historic victory or something like that
so it's uh interesting how we give the
disciplinary nod with that al
on the adjective
um i was just thinking about the
difference like linguistically between
historic and hysteric
like what might what might that be um so
i my question unrelated to historics and

hysterics um really goes back to um you know what patricia was saying about the second law and touches back to what stephen was saying earlier about um the non-existence of negative entropy and and um you know i'm i'm curious about that because i think um you know

a lot of the fep is is um grounded in information dynamics which depends greatly on the existence of negative entropy like the minimization of surprise uh in the terms of information theory so i i was wondering if physics and information theory are are different are they one and the same what are the differences how do we clarify them and similarly um you know one of the claims in the paper is to pay attention to the differences between physics and biology and what you know i've always thought that

the laws of physics are universally applicable and and perhaps they're not but but

um i i think that they should be and you know that this defiance of the second law like are do the laws of physics not apply in biology in biological systems and

um is is it negative entropy that that biological systems do i mean scott david always classically refers to the entropy secretion

of you know biological systems and so i just was wondering how do we reconcile these differences maybe

if you guys have thoughts
yeah great questions stephen you want to
go on that topic or
yeah i think i think there's a really
good
i think this is important because
you see
neg entropy
kind of is a bit like
negative enthalpy in the way it's
formulated i
but the thing is
you have a reduction
only relative
to a starting state starting equilibrium
state in terms of entropy it's it might
have gone down in terms of a starting
state but it's not actually
lower in
like energy is energy right you it can't
be made can't be lost
you know but the entropy
you can have a variation in the entropy
and that variation can be noisy going up
and down you can then extract
information
um you can have things which are more
ordered now that's fair enough you can
have more order but neg entropy
um isn't something which is
a value in that way
it's just a convenient
piece of
temporary
mathematics
to help see whether you're going to have
a force
driving or reducing a reaction

which so that's where the challenge is
because it's you're getting it from the
variations not from
something which is an equilibrium state
because as soon as you say neg entropy
it makes it sound like it's sitting
there with a particular amount of
entropy
that makes sense
um and it's not really like that
so this this is this is like one of the
kind of
misunderstandings i think around
entropy okay interesting um one other
view blue on your
question about information and
thermodynamics
we have some very
slam dunk
info thermo
synthesis
on the processor scale
like the amount of heat that is emitted
or absorbed by certain very very
granular atomic
in multiple senses
informational transitions
and so it's almost like
that's like the the stem
and there's that we know that there is a
connection there
um either
as if
to be sure
if not in actuality so map and territory
information and thermodynamics
potentially coming together
and part of our

curiosity and drive with active
inference
is whether
info dynamics
might apply to higher orders of
organization
and whether we can use some of these
models and ideas to describe
informational patterns and thermodynamic
eventually even bioenergetic
patterns and phenomena
beyond the cpu
and that's where
it's going to be this question of okay
maybe we generalize
and it's useful
but we've lost some reality touch so the
parameter
a temperature parameter on decision
making doesn't refer to
the thermometer reading of the brain
i think we're okay with that so in a way
we are lifting
away from the physical grounding
because we're using terms like
temperature
entropy
in their information and not their
physical
capacity
and as the key questions raised in this
paper um
brought to this discussion
there are opportunities and challenges
with making those kinds of
generalizations it's not just a simple
win
it's things are gained things are lost

some things come into resolution other
things become
incapable of being resolved
so that's like the
some of the excellent questions
that
i think the paper brings to the table
and it combines those questions
along with just
tantalizingly
a look at the history
and the development of
active inference as a framework
and so we can be clear about which
papers said what when
and then where are we headed what would
be our preferences
for
a thermodynamic model of decision making
will we only succeed when it's about the
temperature of the brain
it might be for some person
however other people might just say
if there is a temperature-like parameter
in the sense that that parameter's
formalism looks like temperature
of water in a test tube
then
we have something that's already useful
and interesting
and
that's where
the train ends for me
and then somebody else may continue to
build the rails
off towards wondering whether
is it metaphysically a temperature of
decision making all these other areas

that people can take it
um
and so
it's this trade-off between realism for
generality and utility that we've come
to these
uh meta modeling discussions before
in number 14 and beyond
okay
um
anyone wants to ask a question in the
chat or raise their hand here
otherwise let's see
back to our dot one
questions and see if there's any threads
from dot one
that we want to just um
recap or see if we have a different
perspective on
and then we can just sort of
close
the dot 2 and
jump off into
the next steps so
in
one
we started with some questions again
from a more history of fep and active
about
in what ways has physics
been
used to investigate the fep
we talked a little bit about
how the decosta paper that was 26
bayesian mechanics for stationary
processes
relates to the erogedicity stationarity
discussion

or not we'll see

we talked about then and also a little

bit today

is the fep a theory of life well in the

conclusion it was said to be a

definition

and we talked about some other things it

could be blue

so you still have to unmute you'd think

i i would learn after all this time um

never learn don't change

the um the ergonomy thing i think maybe

you guys discussed that uh before i got

here but but i think like one of the the

points that was made in this paper um

addressing the ergodicity and the fact

that um

you know biological systems are are i

mean i don't think can i don't think

biological systems can be described as

as ergotic and this is a discussion that

that daniel and i have had many times um

i mean sure within a very restricted

space

like you know my body temperature for

example in the range from i mean what

can i survive like 105 fahrenheit

probably and you know maybe 95

fahrenheit so i have like this 10 degree

range and i'm sure at some point in my

lifetime i'll visit all the space within

that range because you can't skip over

you don't go from 96 to 100

and skip any of the points in between so

like you know it's incrementally like of

course i'll visit all of that space um

and so i i do think that that that's an

important question but um yeah that was

really um

addressed by de costa in in that uh 26

live stream um about you know

homeostasis is

a quasi stationary because it's not

perfectly stationary right it's like a

it's a squishy stationary but it's not

you know it doesn't it doesn't sample

the entire we don't sample the entire

state space of say temperature um you

know as a biological system that must

maintain a certain temperature so i

think that that was resolved but did you

guys discuss that before i got here

not the boundedness per se

but

potentially some

[Music]

relevant pieces about arrogancy and

stationarity

um yeah so may i say something sorry

though yeah i have to turn off the video

i'm with my baby here

hello baby um

yeah

um yeah so we didn't we i agree with you

blue i think that's one of the things

that we that we discussed in the paper

so it's very problematic to understand

biological systems such as body systems

uh but uh in the discussion last week

you were saying or maxwell was saying

that

new formulations of the free energy

principle don't

rely on that assumptions that they

dropped that assumption in the new

formulations of the free energy

principle

and what i was saying

before today is that it's still in new

formulations of the free energy

principle

they rely on the existence of weakly

mixing random dynamical systems and to

my understanding mixing systems require

ergodicity anyhow so it's not clear that

they have really dropped the assumption

in the new formulations and if so i

would like to see exactly how the

framework can work without the

assumption of ergodicity i haven't seen

that explicitly i

my belief that that can be done but i

would like to to see it

and

we don't evade requirements just by not

mentioning them

so it's about what papers do mention as

well as reading between lines between

the formalisms between the citations

to really what is the bedrock

or

tautology that the theories rest upon

stephen

you know i think this also might point

to your um cybernetics point daniel um

in the sense of you know where

does something happen at this kind of

ergodic

um

process of mixing and where may it

happen through a

higher level

cybernetic mechanism that the organism

has

that is enabled by
sort of
ergodic
active inference or whatever processes
are
emerging
but the fact that it's emerged from
ergodic processes at multiple nested
levels doesn't necessarily mean that
some of the higher level ways of keeping
something within bounds
then has to
follow that
because if you know you can create a
system
out of all the cells that are doing
their thing so to speak
or being their thing so that we
we this is where it can maybe
um
be that danger that yeah there's an over
extension of a certain process it's like
okay where does
something which is relying on
ergodic processes
or
some other
approaches
able to give way to
another type of um
control theory
yes and over short time horizons
is it possible to have sensory motor
loops that work
whether or not some infinite case
is what it rests upon blue
so patricia i'm curious um to know or to
know what you think about if you've read

the decosta paper about um basinary
basinary processes for
bayesian process bayesian mechanics for
stationary processes there you go um
so i'm curious to know if you've read
that and what you felt was perhaps
lacking
from that
mathematical description
to um meet the requirements that uh you
were referring to for
mixing systems and
ergodicity yeah thank you i i did look
at that but we explicitly wanted to
focus on other assumptions not on the
bayesian assumptions of the paper we
didn't also mention the markov blanket
assumption which is a core assumption in
the free energy principle
and we left that out deliberately first
of all because of lack of space the
paper is long enough second because
there is much more literature on that
than on the other these other
assumptions that we discuss in our paper
so i think in this formulation of um
well i don't i don't have that paper on
the top of my head but i think they they
again uh
rely on weakly mixing dynamical systems
in the paper right
they don't
we can
check they they use that framework
so i think to the extent that they use
that framework if i'm not mistaken then
the same criticisms apply
you need to find phase spaces and you

need the existence of attractors at

least

cool

i just imagine like these networks of

citations like a

some sort of biological metaphor some

swarm of citations with words being

traced between them and connections of

the words and um

i think um we're in the boutique the

artisan the farmer's market phase of

these kinds of analyses and there there

will be a time not so far away

when

these questions can be resolved better

blue

um so yeah just like the paper aside i

mean i i definitely realize that there

are space constraints and also like what

was the timing of your paper versus the

decosta paper and

did you have time to like incorporate

all of those the new like updated um

version of the maths uh so

there's not weekly mixing or mixing

mentioned in the de costa paper and so i

really just would be curious maybe you

want to follow up via email

um to to know what what was missing

from that that formulation in in your

mind for it to be um plausible because

for me too it was very uh

like the the foundation of the fep

existing in in ergodicity or requiring

ergodicity for applicability was very um

jarring like i felt like that's that

can't be right

but um i i felt that it was really

resolved by the decosta paper and so i'm
curious if you um
don't feel the same way
what your hesitation would be so maybe
we want to follow up in in in a dot
three at some point
yeah or maybe or maybe by email so i
would definitely look at this paper and
other new research because that was
published this year right
it's very recent so the papers that we
were
focusing a little bit more on in the
babe in our paper was par it is also
very new and christian 2019 this prop
this free energy principle for particle
physics
uh
in the other papers we look at them uh
very briefly because they were actually
out after our paper was accepted or
maybe during the review process so
it wasn't part of our research
systematic research but we did take a
look at them
uh i think we might have even quote that
paper
but yeah so i i would like to take a
look at that again i'm not sure whether
they use if they use the notion of
information geometry then they probably
also
rely on weekly dynamical systems but
i i would have to look at that again so
we can continue the conversation by
image
counter factual paper spaces you know
you have some sort of clause that gets

triggered

if something happens in the literature

then

this paragraph should be deleted like

pending a future word on this this could

be deleted

um

in the last

little bit

maybe we can just

have any

thoughts uh just

why

does this whole discussion matter for

the fep and for active well the validity

and the adequacy

were how it was described by this paper

why

it

it matters so it's kind of cool because

often

the citations that we discuss they don't

always lay out on the table why it

matters to understand this topic

and uh i think we got that very

clearly from this work so i really

appreciated that and it also laid out

many of the next steps directly not

every next step but

the specific key uh themes and

challenges

are addressed and if there is a obvious

or non-obvious response

that'll be a citation that somebody can

swoop on stephen

yeah i think it's important um because

we are seeing a lot of

interest

a huge amount of interest from in
activism and embodied work and
physicality and how these things can
shape up
and therefore
purely
mathematical arguments
will
limit things i think getting into the
nitty-gritty of the physics which in
itself is
wrestling with some of this because it's
kind of cutting edge for them in terms
of that is really important though i
think because you see some of the
criticisms coming
around in activism
um trying to get arrested on things and
i do think i will say that i think
maybe sometimes saying
go and visit the math
which is ultimately based on ergodicity
which is basically a physics principle
as much as a math principle in terms of
you know how that comes to be present
um
comes up so i think this is this is a
really important
area that physics and
and maybe the sort of the areas of
applied physics start to come into
yes
and we'll just send them on the
runaround okay go to the math and
they'll go to the physics okay now
to the philosophers you go and then
back to another maybe non-academic
department

is that run around the answer what if
there's a stationarity in the runaround
but
very good points um
it's kind of fun because we've had these
questions
for a long time if not
the longest time
yet
many of them
were explicitly addressed by this paper
and by the 31
discussions
so that's kind of cool to see like a
sort of it's not a conclusion on our
series but these
these uh questions which are often the
next step
for other papers
discussed in the dot 2 discussed in the
conclusion or the discussion
that's where this work
picked up
and so that that's a um
a different attribute than we've seen
from some of the other
literature we discussed
um
blue
dave do you have any other thoughts
patricia would you like to give any
final comments
i think i have said enough
unless you have an uh a last question
what papers i i'm a little bit of fun
i'm sorry i'm a little bit of an
outsider in the free energy principle i
must confess but um

i i would like to see how it progressed
so as i as i said at the beginning we
are not
against the the free energy principle a
priori so if our paper can lead to more
research in the free energy principle
direction we would be happy and we would
consider it a success so we are not
necessarily enemies of the free energy
principle and i'm in principle very
interested in the application of
statistical mechanics and even the
language of dynamical systems to biology
so i am in principle interested in this
type of of research and also in the
active inferences interdisciplinary
research
so i would be very interested in see how
this evolved is and if we as
philosophers can contribute to that by
criticizing some points that we find
non-rigorous enough then our job is done
so that's basically our goal
thanks patricia totally agree this um
hopefully we can enact the kind of
dialogue at the micro like a
conversation and a disciplinary
or scholarly dialogue that plays out
over years
and
the philosophers have a very important
role in that blue
so what uh patricia was just saying
reminds me of um you know when we
discussed mike levin's paper in the
summer um
the computational boundary of a self
it's

it's you know there's the
instrumentalist view
and then the realist view
and then the utility axis in that in
that uh diagram and so definitely um i
thought that this paper was useful in
provoking dialogue and um
really kind of uh
i would like to leverage your vantage
point to kind of grind into some of the
deeper maybe more mathematical stuff
that's kind of beyond me because it's
not really my my background but to kind
of you know dig into that hole and make
sure that there's a solid foundation
under the hole or is it a hole that
pokes all the way through and leads to
nothing on the other end so i think um
you know it's useful for for getting at
those those questions
my last question patricia would be what
areas of literature or papers might you
recommend that we
dive into
so that we can have a comprehensive view
and um
know where we're coming from and where
we're going
uh
i have i can mention many many babies so
you can do that better by email again so
if you're interested in literature and
some specific
uh
research or topic just send me an email
and i can recommend philosophically to
you but i would like to especially
recommend one paper so if i can choose

and i don't know the title right
hopefully but it's
it's a paper by karim thibault uh
who is called something about
imperialism and migration so i can look
at that now
because it really
it's a very general paper in philosophy
of science
where drastic
they draw a very nice distinction
between migration of models and
imperialism of models that i think will
be also useful in the interdisciplinary
research that you are doing
uh because they they stress the
difference about using the methods for
or exporting the methods from one
discipline to the other
uh
in a non um imperialist way which would
imply that in the new discipline you
have to justify
the model you have to justify the the
the mathematical methods that's that's
the one models on the move migration and
parallelism brilliant
and they distinguish between that that
they call migration of models from
imperial imperialism in which you just
extrapolate the same models from one
discipline to the other and you try to
expand your discipline which would be
sort of the
the attitude that you were
you were um
that was illustrated in the in the
comics that you were discussing last

last week so i think that paper is
especially useful for for for people
doing interdisciplinary research because
it's a very useful
distinction this model is on the move
thanks for the awesome suggestion so
just to the point about
emails or however it goes just consider
the dialogue opened we really appreciate
that both of you came to discuss and
also interacted with the dot zero and
dot one videos so just
awesome work with uh research and thanks
again for
helping our
lab do what it does
and this is also an excellent suggestion
because it's a provocative
title
but we're talking about
provocative ideas
and with broad
scope and this kind of takes that map
territory discussion
to a new
place
potentially talking about
movements
and imperialism and
colonization all these other things that
come up in different terms but we've
noted this in our upcoming streams
calendar
so after our lab meeting next week
we'll have stochastic chaos and markov
blankets so that will complement some of
the things that we didn't talk about in

31.

we'll talk in 33 thinking like a state
with aval aka serval
and then we have two more slots open so
perhaps we'll select one of these that
we've already listed
or perhaps
somebody wants to join if an author
wants to discuss
and um
other than that
i think that brings us to a close today
so again big appreciation to the authors
for their work and engagement with the
lab and thanks steven dave and blue and
everyone else who participated in
maxwell for joining the dot one so
good times and we will see you next time
thank you thank you very much bye